Interpreting the recent rise in invasive pneumococcal disease (IPD) in Belgium:

The role of surveillance coverage and diagnostic activity

C. Dambre[1], L. Cuypers[2,3], S. Desmet[2,3]

Background

Marked rise in IPD cases since 2022, with record numbers in 2024 and 2025.

As surveillance by the National Reference Center is not exhaustive, changes in its coverage or in diagnostic practices could influence observed case numbers and trends.

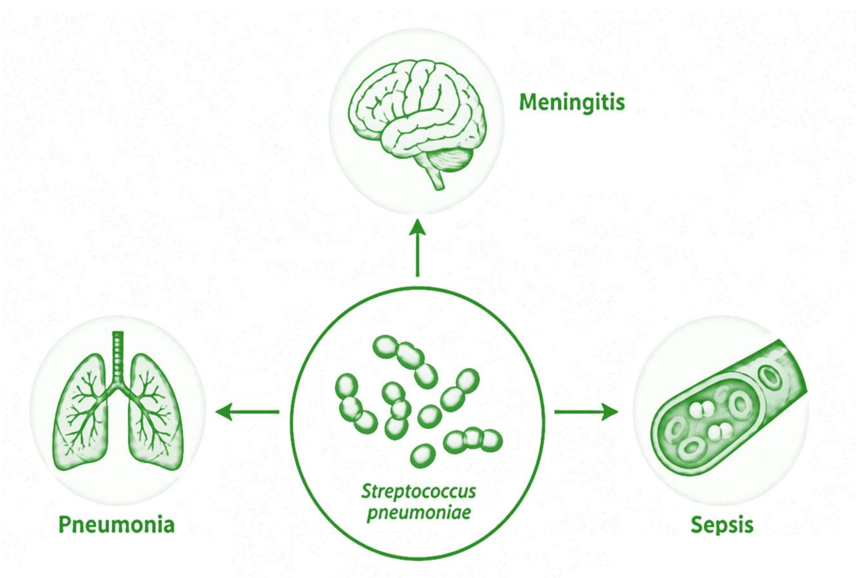


Fig. clinical manifestations of invasive pneumococcal disease

OBJECTIVE

To assess if this is a real epidemiological increase or a surveillance artefact

Methods

Data sources:

- RIZIV reimbursement data for blood cultures tests ~ proxy for IPD diagnosis
- NRC IPD surveillance (voluntary, laboratory-based): blood culture isolates selected for analyses

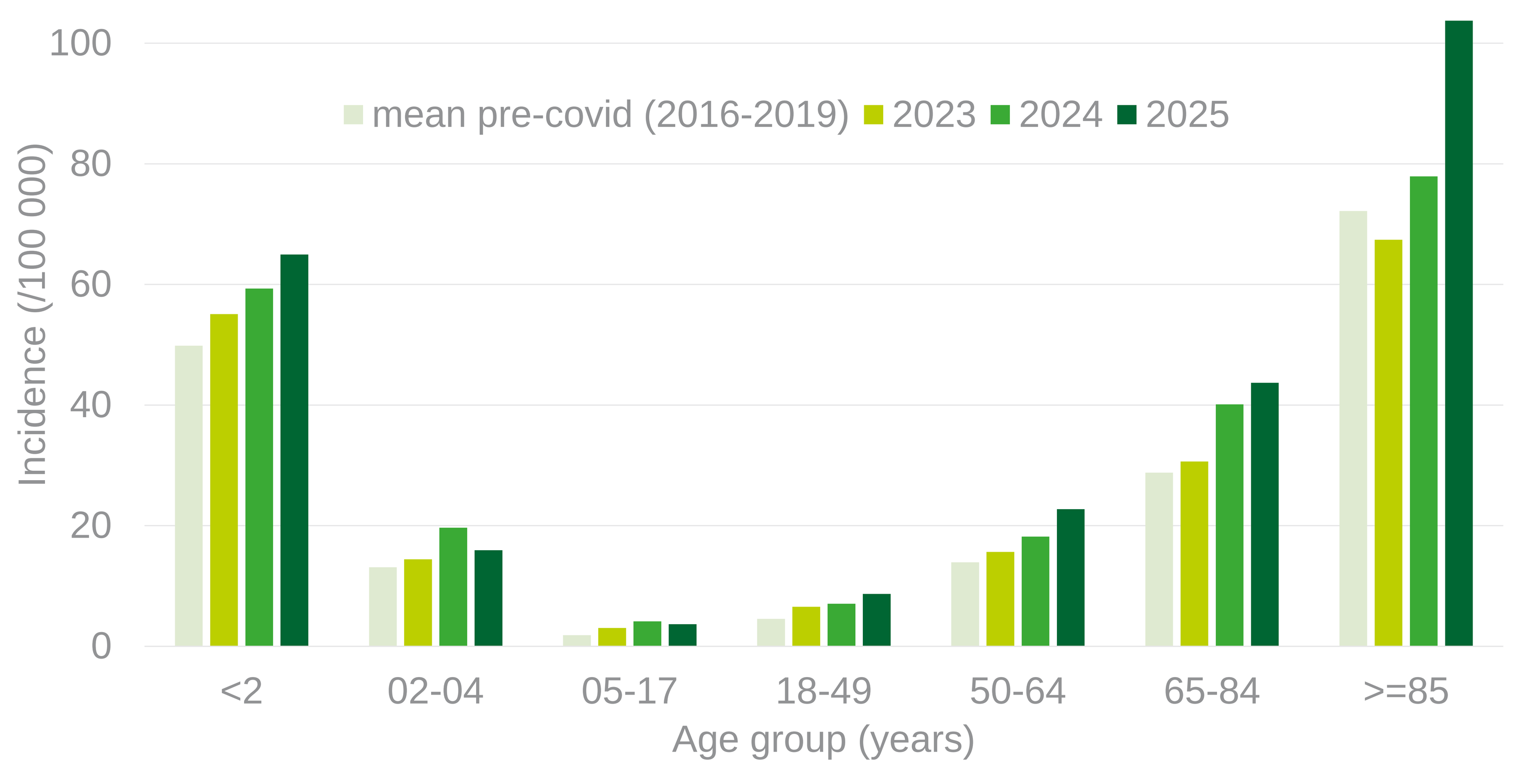
Study period: 2016-2024

Indicators:

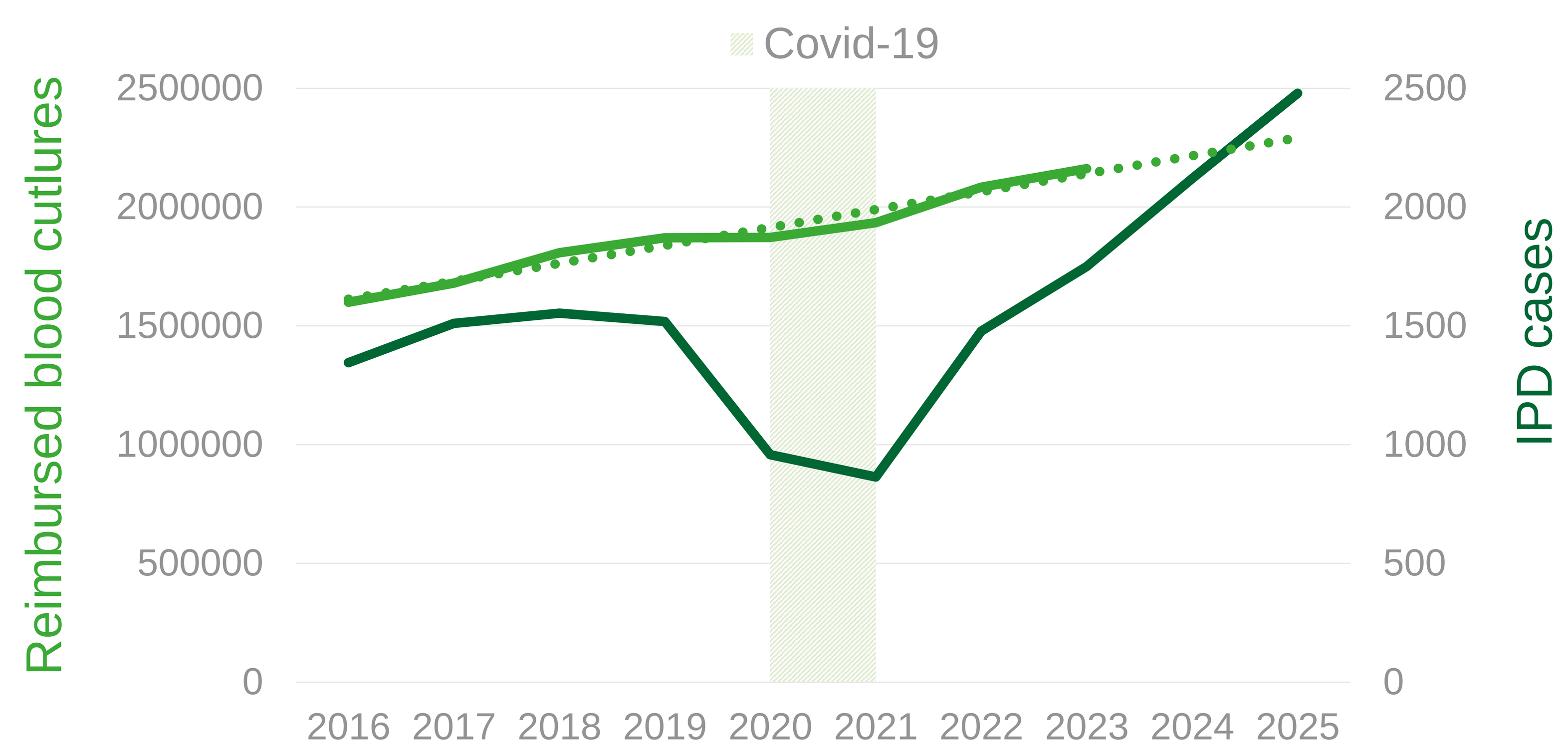
- IPD incidence: age-specific incidence per 100 000 population
- Surveillance coverage: blood culture tests from labs participating to NRC surveillance / total blood cultures
- Diagnostic activity: number of IPD cases / total blood cultures

Results

IPD incidence by age group, Belgium

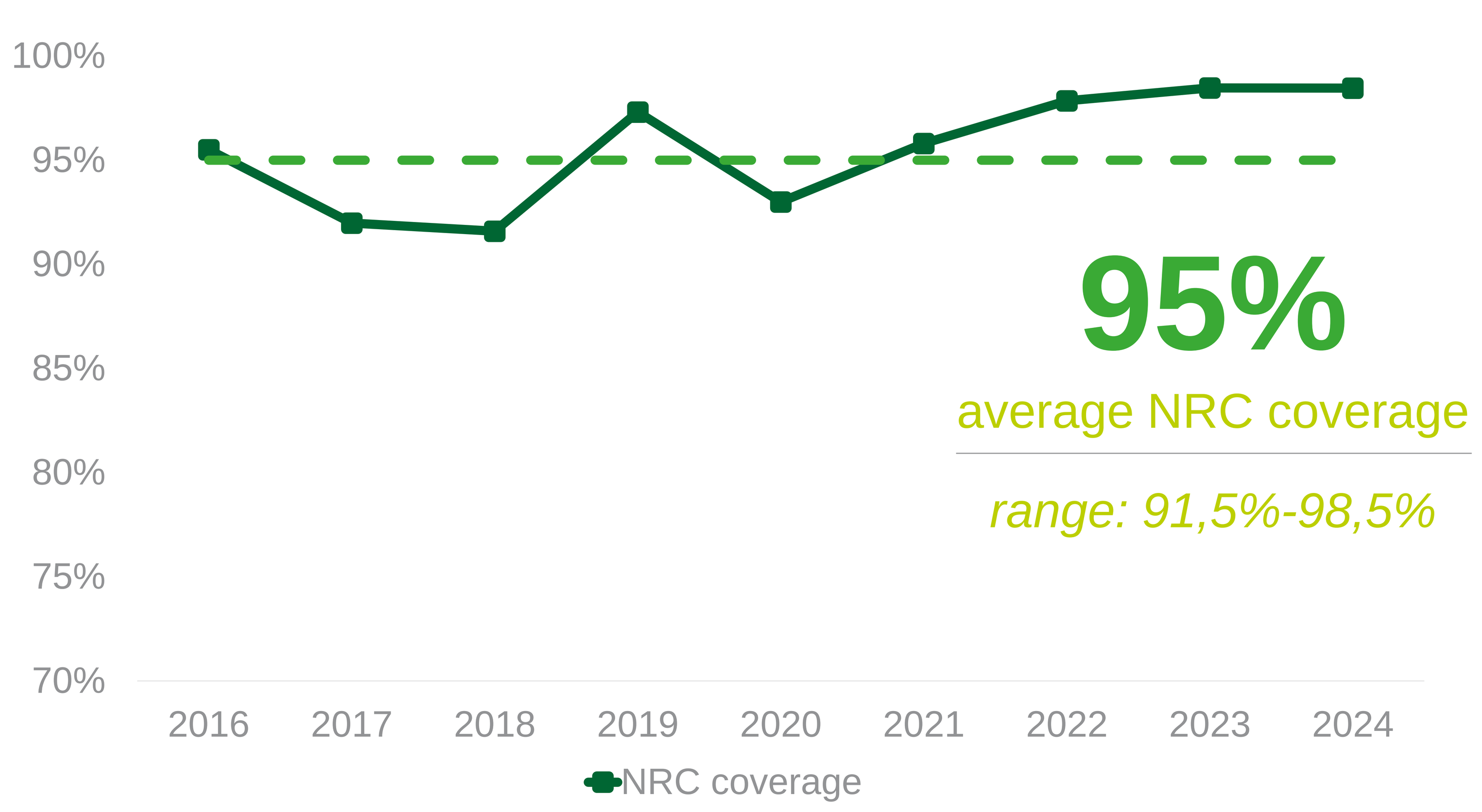


Diagnostic and IPD trends over time, Belgium



RIZIV data for 2024 and 2025 not yet available. Dotted trend line for illustration only.

NRC surveillance coverage over time



Key findings

- Record case numbers in 2025 (n=2479)
- Number of blood cultures increases steadily over time likely contributing to higher case ascertainment.
- Sharp increase since 2022 occurs alongside only a modest rise in tests, and stable positivity ratios.
- ~95% of blood cultures were performed by laboratories participating to NRC-surveillance.
- Coverage exceeded 97% in recent years, likely due to lab mergers rather than expansion of the network.

NRC coverage has been consistently high. While increased diagnostics may partly contribute to higher case detection, it alone does not explain the recent observed surge. **The rise in IPD since 2022 most likely reflects a true epidemiological increase**.

1 Epidemiology and Public Health, Sciensano, Brussels, Belgium
2 Department of Laboratory Medicine, National Reference Centre for Invasive Pneumococci, University Hospitals Leuven, Leuven, Belgium
3 Department of Microbiology, Immunology and Transplantation, Laboratory of Clinical Microbiology, KU Leuven, Leuven, Belgium

Cato.Dambre@sciensano.be

sciensano